

# Control of Nutrient Stress-Induced Metabolic Reprogramming by PKC in Tumorigenesis

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## Evidence abstract

Tumor cells have high-energetic and anabolic needs and are known to adapt their metabolism to be able to survive and keep proliferating under conditions of nutrient stress.

We show that PKC deficiency promotes the plasticity necessary for cancer cells to reprogram their metabolism to utilize glutamine through the serine biosynthetic pathway in the absence of glucose.

PKC represses the expression of two key enzymes of the pathway, PHGDH and PSAT1, and phosphorylates PHGDH at key residues to inhibit its enzymatic activity.

Interestingly, the loss of PKC in mice results in enhanced intestinal tumorigenesis and increased levels of these two metabolic enzymes, whereas patients with low levels of PKC have a poor prognosis.

Furthermore, PKC and caspase-3 activities are correlated with PHGDH levels in human intestinal tumors.

Taken together, this demonstrates that PKC is a critical metabolic tumor suppressor in mouse and human cancer.

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## Evidence checklist

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2. We show that PKC deficiency promotes the plasticity necessary for cancer cells to reprogram their metabolism to utilize glutamine through the serine biosynthetic pathway in the absence of glucose.
3. PKC represses the expression of two key enzymes of the pathway, PHGDH and PSAT1, and phosphorylates PHGDH at key residues to inhibit its enzymatic activity.
4. Interestingly, the loss of PKC in mice results in enhanced intestinal tumorigenesis and increased levels of these two metabolic enzymes, whereas patients with low levels of PKC have a poor prognosis.
5. Furthermore, PKC and caspase-3 activities are correlated with PHGDH levels in human intestinal tumors.
6. Taken together, this demonstrates that PKC is a critical metabolic tumor suppressor in mouse and human cancer.

## Source continuity

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